

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green. They are positioned diagonally, with the blue one partially covering the green one.

Final Project Student Designed Lab: Physical Model for a Spring Pendulum

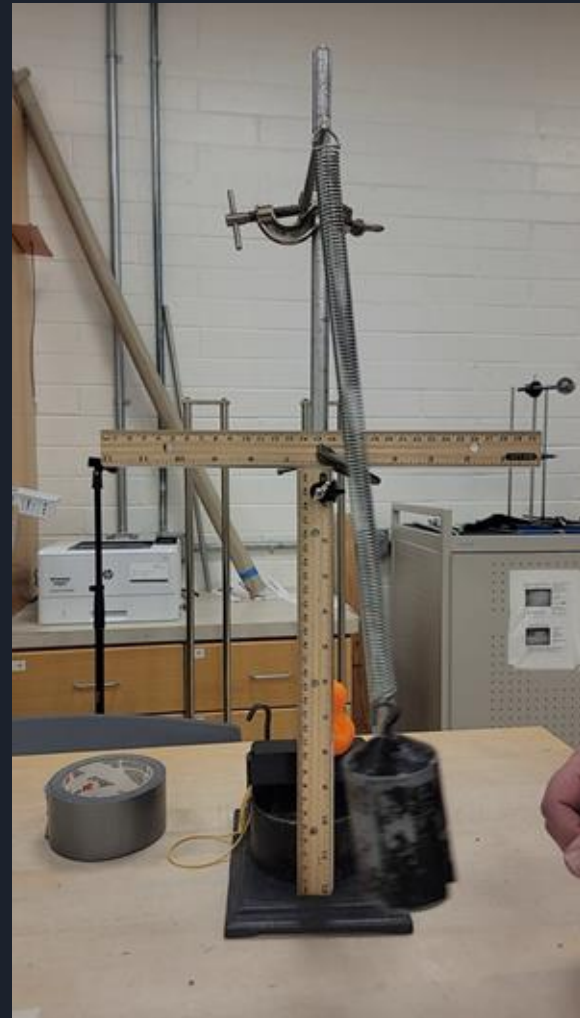
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Method of Collecting Data

Made set-up with form of measurement (rulers connected)

Video recorded with 15 fps and collected points of location of mass for each frame

Measured approximate measurements (m,l,k,h) for starting variables of gradient code





Challenges

- Measuring position of mass: (protractor didn't work well)
- Video analysis: how to log data
- Code: many local minima, troublesome fit equations

Approximations made:

- Small angle
- Used only 2 dimensions, there may have been slight movement in the third dimension
- Damping
- Axis friction
- Massless spring

Finding an accurate fit model

Lagrange equations:

2 degrees of freedom - r (length of the spring) & Φ (angle of pendulum from vertical)

$r = l$ (equilibrium length of spring) + ϵ (radial change from spring equilibrium)

After small angle approximations:

$$\frac{\delta \mathcal{L}}{\delta \phi} = \frac{d}{dt} \frac{\delta \mathcal{L}}{\delta \dot{\phi}}$$



$$-mgr \sin \phi = mr^2 \ddot{\phi} + 2mr \dot{r} \dot{\phi}$$

$$\frac{\delta \mathcal{L}}{\delta r} = \frac{d}{dt} \frac{\delta \mathcal{L}}{\delta \dot{r}}$$



$$mr \dot{\phi}^2 + mg \cos \phi - kr = m \ddot{r}$$

$$-mg(l + \epsilon) \sin \phi = m(l + \epsilon)^2 \ddot{\phi} + 2m(l + \epsilon)(\dot{l} + \dot{\epsilon}) \dot{\phi}$$

$$m(l + \epsilon) \dot{\phi}^2 + mg \cos \phi - k(l + \epsilon) = m(\ddot{l} + \ddot{\epsilon})$$

$$\phi(t) = A \cos(\sqrt{\frac{g}{l}} t - \delta)$$

$$\epsilon(t) = B \cos(\sqrt{\frac{k}{m}} t) + C \sin(\sqrt{\frac{k}{m}} t)$$



Method of minimizing χ^2 / Optimizing fit variables

Gradient Search Method

New fit equations

Lots of local minima

Fitting parabola to the last three points

$$\phi(t) = \left(A - B \cos(Ct) \right) \cos \left(\sqrt{\frac{g}{l}} t - \delta \right)$$

$$\epsilon(t) = \left(A - B \cos(Ct) \right) e^{-\beta t} \cos \left(\sqrt{\frac{k}{m}} t - \delta \right)$$

Results

Φ Equation: $A = -0.125 \pm 1.586e-10$, $C = 1.900 \pm 9.518e-09$, $I = 0.325 \pm 6.345e-11$, $\delta: 0.098 \pm 3.173e-10$

$B: 0.260 \pm 6.345e-10$ (probably not valid)

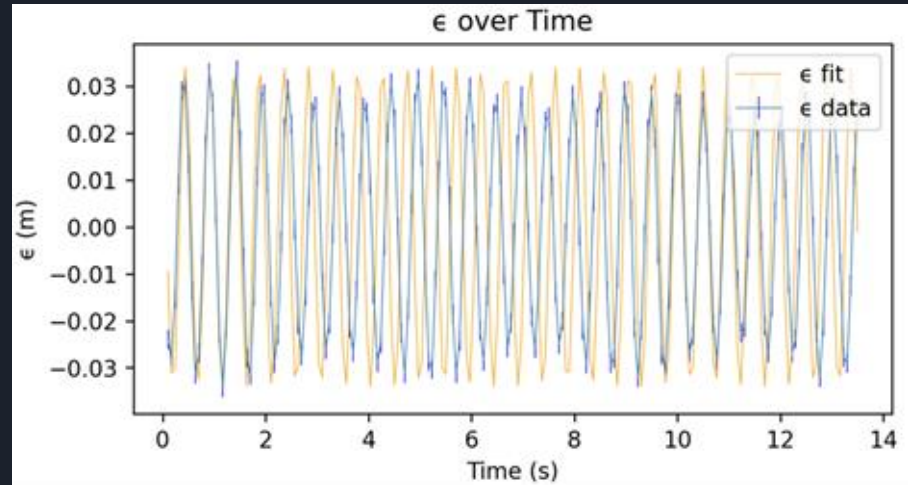
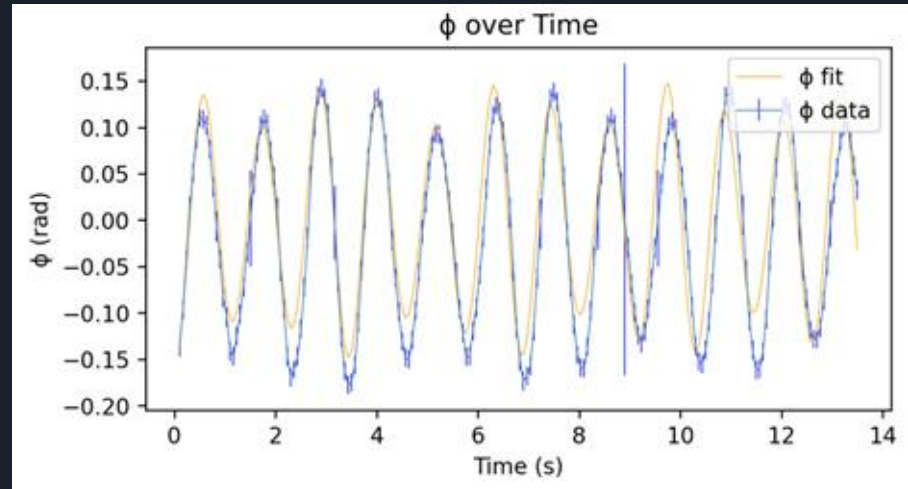
Measured $I = 0.261 \pm 0.002$ m

Chi Squared: 13.825, Correlation Coefficient: 0.923

ϵ Equation: $B = 0.001 \pm 9.463e-12$, $C = 0.030 \pm 4.731e-11$, $\beta = 4.380e-5 \pm 2.366e-14$

$A = 0.040 \pm 4.731e-11$, $\delta = -0.400 \pm 9.463e-10$
(probably not valid)

Chi Squared: 51.386, Correlation Coefficient: 0.175





Future Experiments

Better measuring equipment

Optimization technique that handles local minima better

What happened with B parameter (phi equation), A and δ (epsilon equation)? Fix code accordingly

Physical meaning for other coefficients (whether they have to do with friction/damping, etc.)